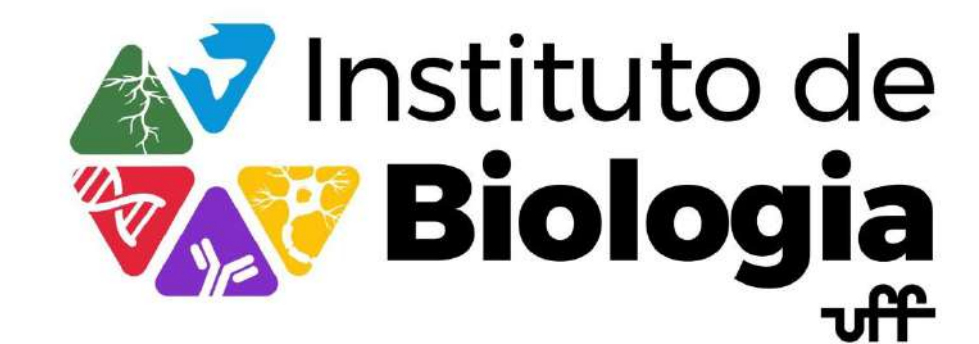


Radiometric analysis of sediments and water from the Lagoons of East Fluminense, Rio de Janeiro.



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1. INTRODUCTION

As a complement to the analysis of aquatic organisms used for food, it is also necessary to consider the concentration of radionuclides in sediments, since soil tends to accumulate radionuclides originating from natural processes (e.g., erosion, leaching, and surface runoff) (Jibiri et al., 2023). These naturally occurring radioactive materials (NORM), such as uranium, thorium, and potassium-40, originate from the geological composition of a region and are a significant source of environmental radiation. Radionuclides enter aquatic ecosystems and are absorbed by fish and crustaceans, leading to bioaccumulation and transfer through the food chain to humans. This exposure to ionizing radiation can cause adverse biological effects. Therefore, monitoring environmental radioactivity is crucial for assessing health risks and protecting ecosystems, serving as a vital reference for detecting changes caused by human activities.

2. MATERIAL AND METHODS

2.1 STUDY AREA

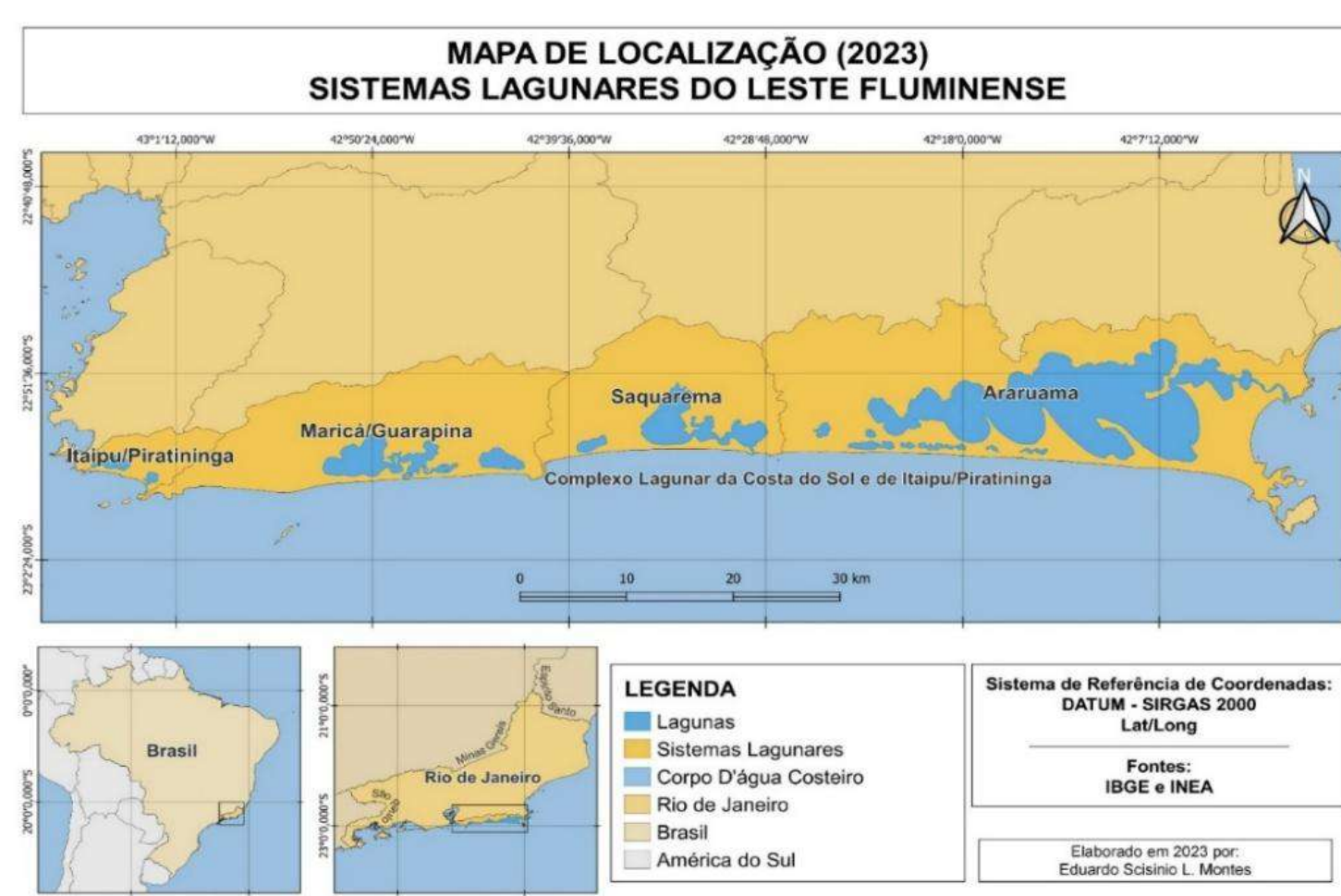


Figure 1: Location map of the Lagoons of the East Fluminense Lagoon System.

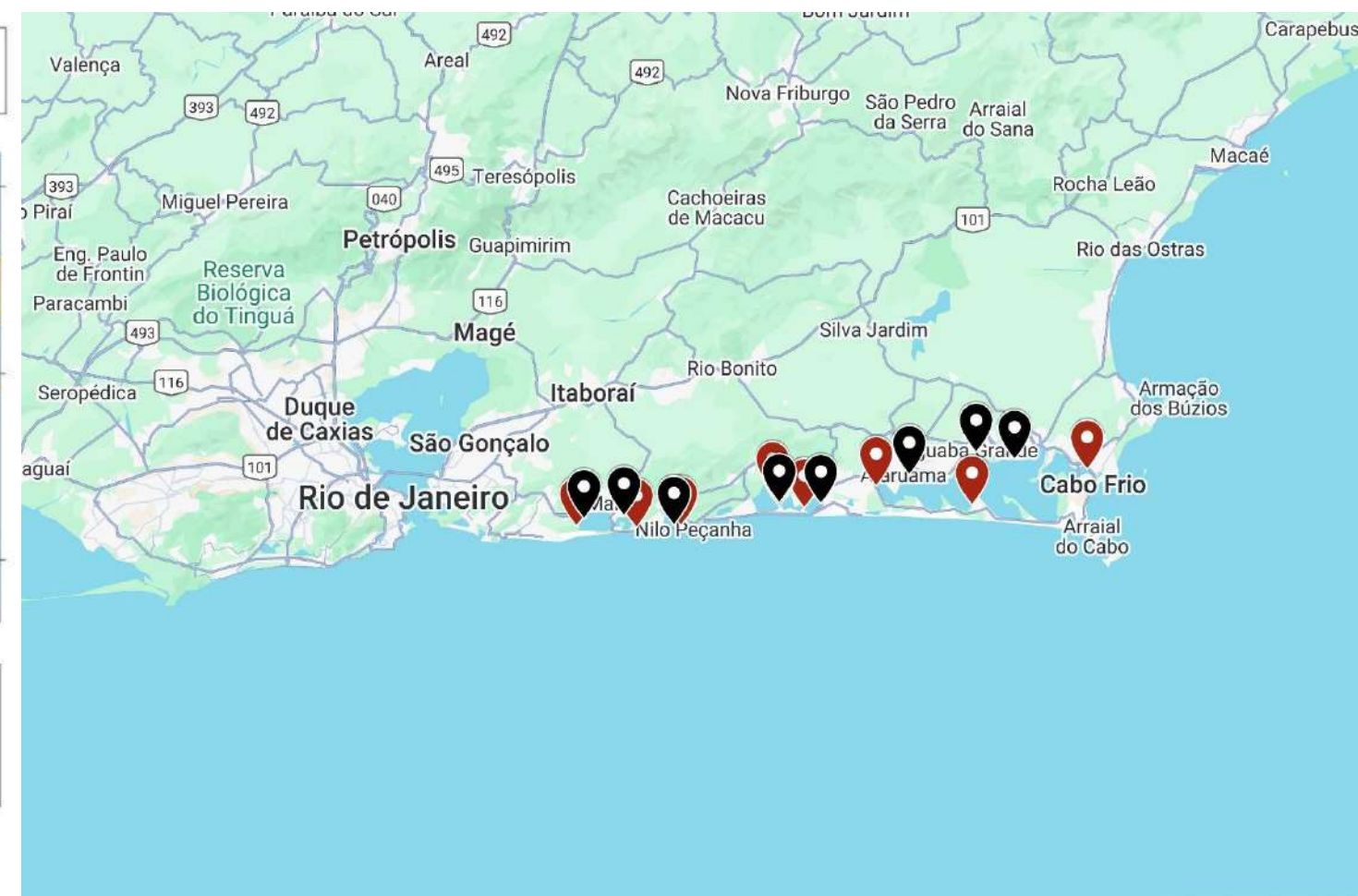


Figure 2: Sampling points in the lagoons of Maricá, Saquarema and Araruama.

2.2 SAMPLING, ANALYSIS AND SAMPLE PREPARATION



3. RESULTS AND DISCUSSIONS

Sediment August					
	²²⁶ Ra	²²⁸ Ra	²³⁸ U	²³² Th	⁴⁰ K
Maricá					
Point 1	11.12 ± 0.87	19.70 ± 1.58	7.42 ± 1.06	6.93 ± 0.67	6.03 ± 1.76
Point 2	8.27 ± 0.78	6.3 ± 1.52	3.9 ± 1.02	4.75 ± 0.46	14.74 ± 2.17
Point 3	1.8 ± 0.25	0.27 ± 0.50	0.69 ± 0.35	0.96 ± 0.13	5.66 ± 0.78
Saquarema					
Point 1	9.94 ± 0.82	15.49 ± 1.56	4.98 ± 1.02	7.82 ± 0.53	29.85 ± 2.60
Point 2	6.37 ± 0.66	3.07 ± 1.26	3.54 ± 0.94	5.22 ± 0.35	7.18 ± 1.66
Point 3	44.91 ± 2.17	44.17 ± 2.89	23.41 ± 1.80	27.64 ± 1.22	177.23 ± 8.28
Araruama					
Point 1	4.92 ± 0.06	2.77 ± 1.69	2.02 ± 1.19	3.13 ± 1.11	31.11 ± 12.64
Point 2	8.48 ± 0.69	13.32 ± 1.32	5.1 ± 0.82	5.99 ± 0.48	4.68 ± 1.46
Point 3	5.9 ± 0.71	4.92 ± 1.44	4.1 ± 0.99	3.72 ± 0.41	39.9 ± 2.95
Sediment September					
	²²⁶ Ra	²²⁸ Ra	²³⁸ U	²³² Th	⁴⁰ K
Maricá					
Point 1	16.66 ± 1.02	27.95 ± 2.04	8.65 ± 1.11	10.63 ± 0.84	19.25 ± 2.08
Point 2	17.77 ± 1.11	22.34 ± 1.87	8.07 ± 1.15	13.64 ± 0.69	29.11 ± 2.63
Point 3	10.32 ± 0.70	19.16 ± 1.42	4.99 ± 0.75	7.48 ± 0.59	14.39 ± 1.54
Saquarema					
Point 1	10.61 ± 0.82	10.67 ± 1.46	4.26 ± 0.95	5.81 ± 0.46	24.45 ± 10.75
Point 2	3.65 ± 0.50	1.88 ± 1.01	1.57 ± 0.69	2.03 ± 0.27	3.66 ± 1.36
Point 3	8.86 ± 0.62	4.54 ± 1.07	3.63 ± 0.68	5.74 ± 0.32	85.78 ± 4.26
Araruama					
Point 1	4.02 ± 0.55	3.76 ± 1.15	3.08 ± 0.72	2.07 ± 0.32	169.63 ± 7.67
Point 2	2.52 ± 0.41	1.28 ± 0.88	0.96 ± 0.57	1.92 ± 0.23	75.48 ± 3.84
Point 3	4.01 ± 0.58	2.54 ± 1.21	2.67 ± 0.76	2.49 ± 0.32	117.29 ± 5.69
Water August					
	²²⁶ Ra	²²⁸ Ra	²³⁸ U	²³² Th	⁴⁰ K
Maricá					
Ponto 1:	< MDA	1.58 ± 25.38	1.31 ± 21.06	1.19 ± 17.02	3.01 ± 48.33
Ponto 2:	< MDA	< MDA	1.04 ± 15.45	1.05 ± 8.53	4.56 ± 67.31
Ponto 3:	0.24 ± 3.80	0.37 ± 5.75	0.04 ± 0.72	< MDA	1.66 ± 25.66
Saquarema					
Ponto 1:	< MDA	< MDA	< MDA	< MDA	2.54 ± 37.02
Ponto 2:	0.02 ± 0.53	1.49 ± 2.46	0.38 ± 1.02	< MDA	3.76 ± 5.87
Ponto 3:	< MDA	< MDA	0.27 ± 4.32	< MDA	2.51 ± 39.21
Araruama					
Ponto 1:	< MDA	0.04 ± 1.26	0.09 ± 1.60	< MDA	6.78 ± 102.83
Ponto 2:	< MDA	0.2 ± 2.95	0.12 ± 1.70	< MDA	3.19 ± 45.57
Ponto 3:	< MDA	0.28 ± 4.19	0.74 ± 10.80	0.54 ± 3.44	2.5 ± 36.72

Table 1: Activity concentration, in Bq kg⁻¹, of radionuclides found in sediment and water in August and September (MDA: Minimum Detectable Activity).

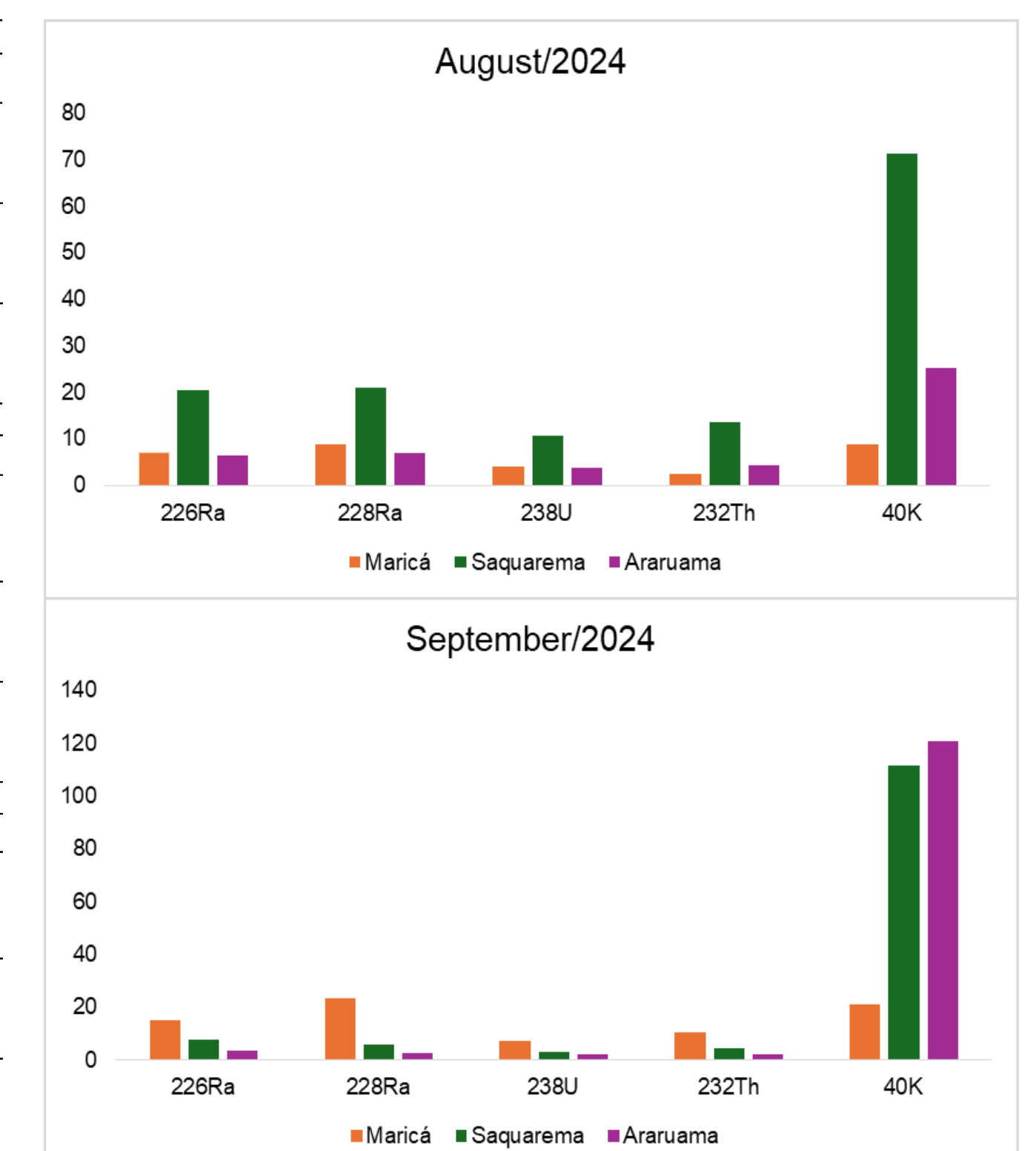


Figure 5: Graphs comparing the average concentration of Bq kg⁻¹ found in the sediments of the lagoons in August and September.

Concentration in sediment (Bq kg ⁻¹)					
	²²⁶ Ra	²²⁸ Ra	²³⁸ U	²³² Th	⁴⁰ K
UNSCEAR 2000					540 – 750
Povinec et. al, 2007	2,5 ± 0,6	1,5 ± 0,5	35,3 ± 1,7*	0,018 ± 0,008	6,8 ± 0,4*
Uosif 2007	20,0 ± 2,5	24,8 ± 1,5		183,9 ± 9,2	
Santawamaitre et. al, 2014			29,19 ± 0,10	30,43 ± 0,10	309,0 ± 1,6
Noli & Tsamos (2018)			19,2–28,7	11,0–38,7	183,0–585
			9,28–24,2	0,43–12,7	1,11–145
Bramha, et al. (2019)	26,27			81,61	620,09
Concentration in water					
	²²⁶ Ra	²²⁸ Ra	²³⁸ U	²³² Th	⁴⁰ K
Noli & Tsamos (2018)			26,55 ± 1,49		
			45,1 ± 2,75		
			56,2 ± 3,63		
			61,78 ± 4,49		

Table 2: Values found in other studies for sediments, soils from lagoons, lakes and rivers.

4. CONCLUSIONS

This study analyzed natural radionuclides (²²⁶Ra, ²²⁸Ra, ²³²Th, ²³⁸U, ⁴⁰K) in sediments and waters from coastal lagoons in eastern Rio de Janeiro (Maricá-Guarapina, Saquarema, Araruama). Sediment concentrations were within reference ranges but varied significantly, with highest levels in Saquarema, potentially linked to low sanitation coverage (27.36% sewage access) and anthropogenic inputs. Water concentrations were below detection limits, consistent with radionuclide adsorption to sediments. Results align with typical coastal ecosystems but highlight region-specific variations due to geological and land-use factors. These findings underscore the need for improved environmental management and policies to reduce contamination. Future work should include long-term monitoring, bioaccumulation studies in organisms, and detailed interaction mechanisms in increasingly urbanized watersheds.

5. ACKNOWLEDGMENTS

6. REFERENCES

